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0178/03

October/November 2015

2 hours

Candidates answer on the Question Paper.

Additional Materials: Electronic Calculator
Geometrical Instruments
Tracing Paper (optional)

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES.

Answer all questions.

If working is needed for any question it must be shown below that question.

Electronic calculators should be used.

If the degree of accuracy is not specified in the question, and if the answer is not exact, give the answer to three significant figures. Give answers in degrees to one decimal place.

For π use either your calculator value or 3.142.

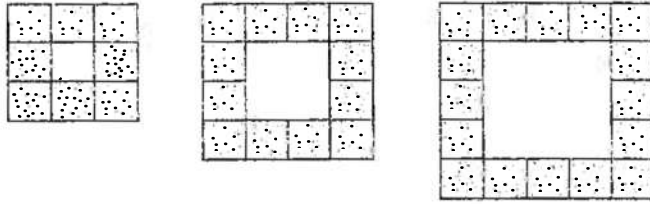
The number of marks is given in brackets [] at the end of each question or part question.

The total of the marks for this paper is 100.

This document consists of 15 printed pages and 1 blank page.



- 1 Consider the sequence of square photos surrounded by squares (shaded in diagrams).



- (a) Draw the next diagram in the sequence.

[1]

- (b) How many shaded squares are there in diagram 7?

Answer (b) [1]

- (c) Complete the table below.

Diagram number	1	2	3	4	7	11	n
Number of shaded squares	8	12	16				
Area of the photo	1	4	9				

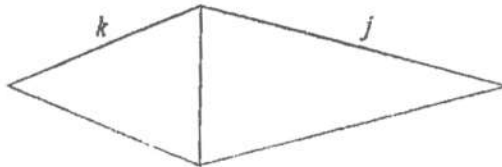
[5]

- 2 (a) Solve the equation.

$$3x - 4 = 14$$

Answer (a) $x = \dots\dots\dots$ [2]

- (b) A kite has sides of length j cm and k cm.



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- (i) Write down an expression, in terms of j and k , for the perimeter of the kite.

Answer (b)(i) $\dots\dots\dots$ cm [1]

- (ii) The perimeter of the kite is 72 cm.

Write down an equation in j and k .

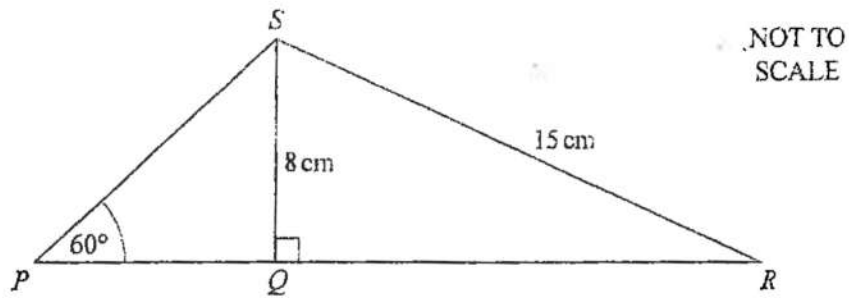
Answer (b)(ii) $\dots\dots\dots$ [1]

- (iii) Find the value of k when $2j = 8k$.

Answer (b)(iii) $k = \dots\dots\dots$ [2]



- 3 In the diagram, PQR is a straight line.
Angle $QPS = 60^\circ$, $QS = 8\text{ cm}$ and $RS = 15\text{ cm}$.



Calculate

- (a) the length QR ,

Answer (a) cm [3]

- (b) the angle QRS ,

Answer (b) [4]

- (c) the angle PSR ,

Answer (c) [1]

- (d) the length PQ .

Answer (d) cm [3]

- 4 (a) A bus travelling from Maseru to Mokhotlong leaves Maseru at 06 15.
The bus takes 2 hours and 30 minutes from Maseru to Botha-Bothe.

Find the time that the bus arrives at Botha-Bothe.

Answer (a) [1]

- (b) The distance from Maseru to Botha-Bothe is 130 km.

Find the average speed of the bus from Maseru to Botha-Bothe.

Answer (b) km/h [3]

- (c) The bus leaves Botha-Bothe at 09 35.

How long does the bus stop at Botha-Bothe?

Answer (c) min [2]

- (d) The bus travels at an average speed of 45 km/h from Botha-Bothe to Mokhotlong.
It arrives at Mokhotlong at 13 35.

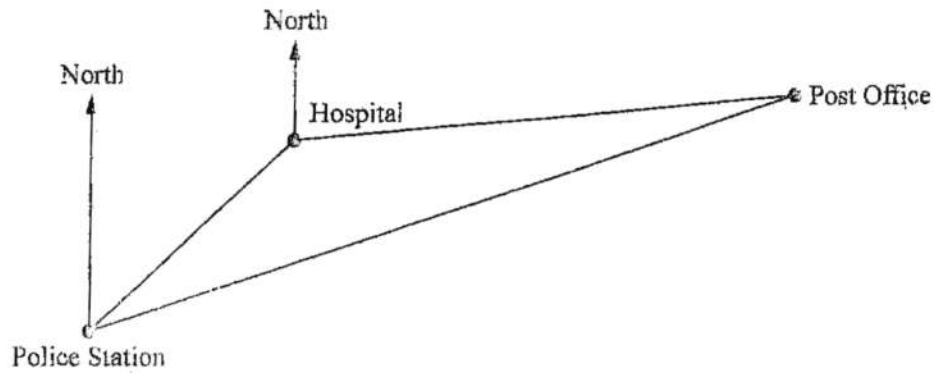
Find the distance from Botha-Bothe to Mokhotlong.

Answer (d) km [3]



- 5 The diagram shows the position of each of three buildings in a town.

The scale is 1 cm represents 50 m.



- (a) Measure and write the bearing of the hospital from the police station

Answer (a) [2]

- (b) Find the actual distance of the post office from the police station.

Answer (b) m [2]

- (c) A supermarket is to be constructed such that it is

- I equidistant from the police station and the post office and
- II 300 m from the hospital.

Construct and mark the position of the supermarket on the diagram.

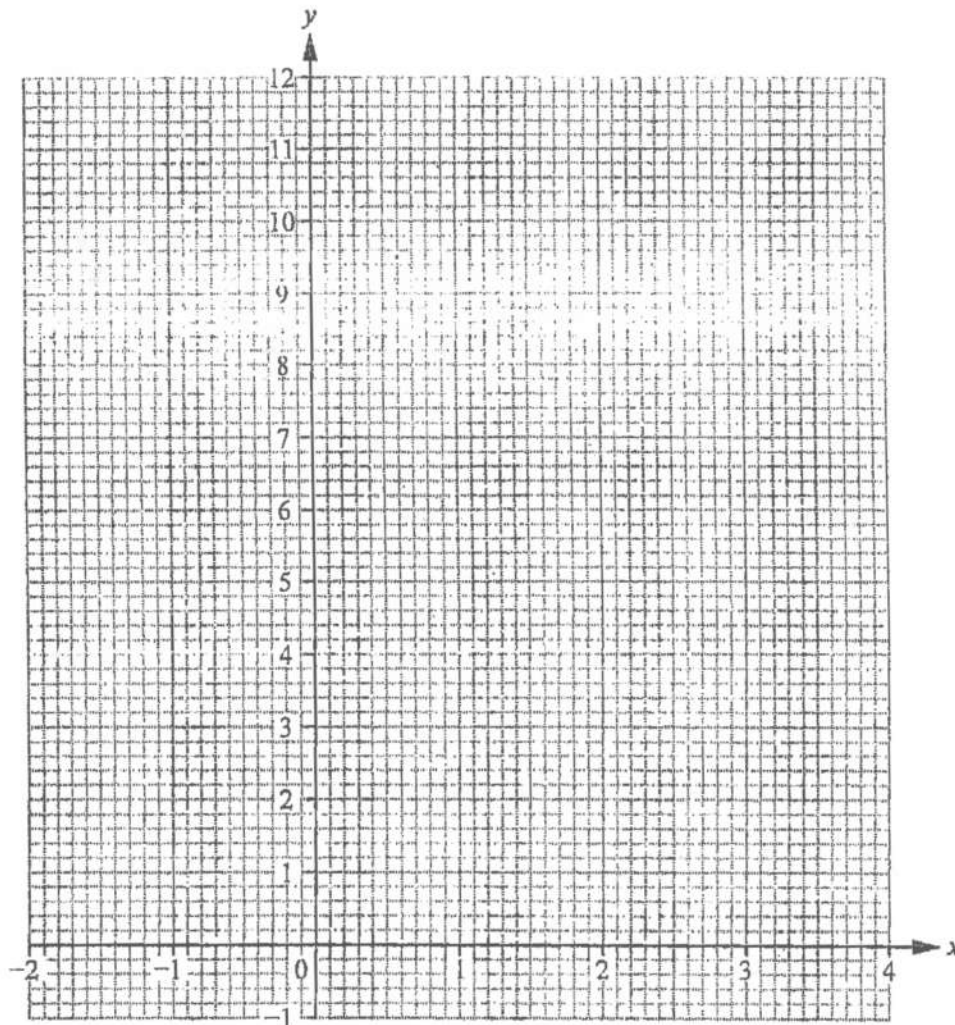
[5]

- 6 (a) Complete the table for the values of $y = x^2 - 2x + 3$.

x	-2	-1	0	1	2	3	4
y	11		3		3	6	

[3]

- (b) On the grid, draw the graph of $y = x^2 - 2x + 3$ for $-2 \leq x \leq 4$.



[3]

- (c) For the drawn graph, write the equation of the line of symmetry.

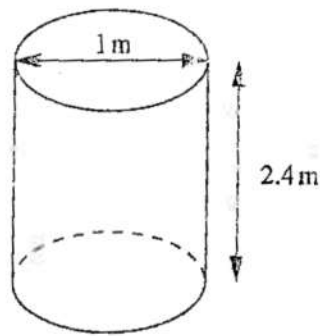
Answer (c) [1]

- (d) Use your graph to solve the equation $x^2 - 2x + 3 = 7$.

Answer (d) $x = \dots\dots\dots$ or $x = \dots\dots\dots$ [2]



- 7 The diagram shows a cylindrical water tank.
The tank has a diameter of 1 m and height of 2.4 m.



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- (a) The curved part of the cylinder is made from a rectangular sheet as shown in the diagram.



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Calculate

- (i) l , the length of the sheet.

Answer (a)(i) m [2]

- (ii) the area of the sheet.

Answer (a)(ii) m^2 [2]

- (b) One litre of paint covers 2 m^2 .

Calculate the number of one litre cans of paint needed to cover one side of the sheet.

Answer (b) [2]

- (c) (i) Calculate the volume of water that can fill the tank.

Answer (c)(i) m^3 [2]

- (ii) The empty tank is filled with water at the rate of $0.02 \text{ m}^3/\text{s}$.

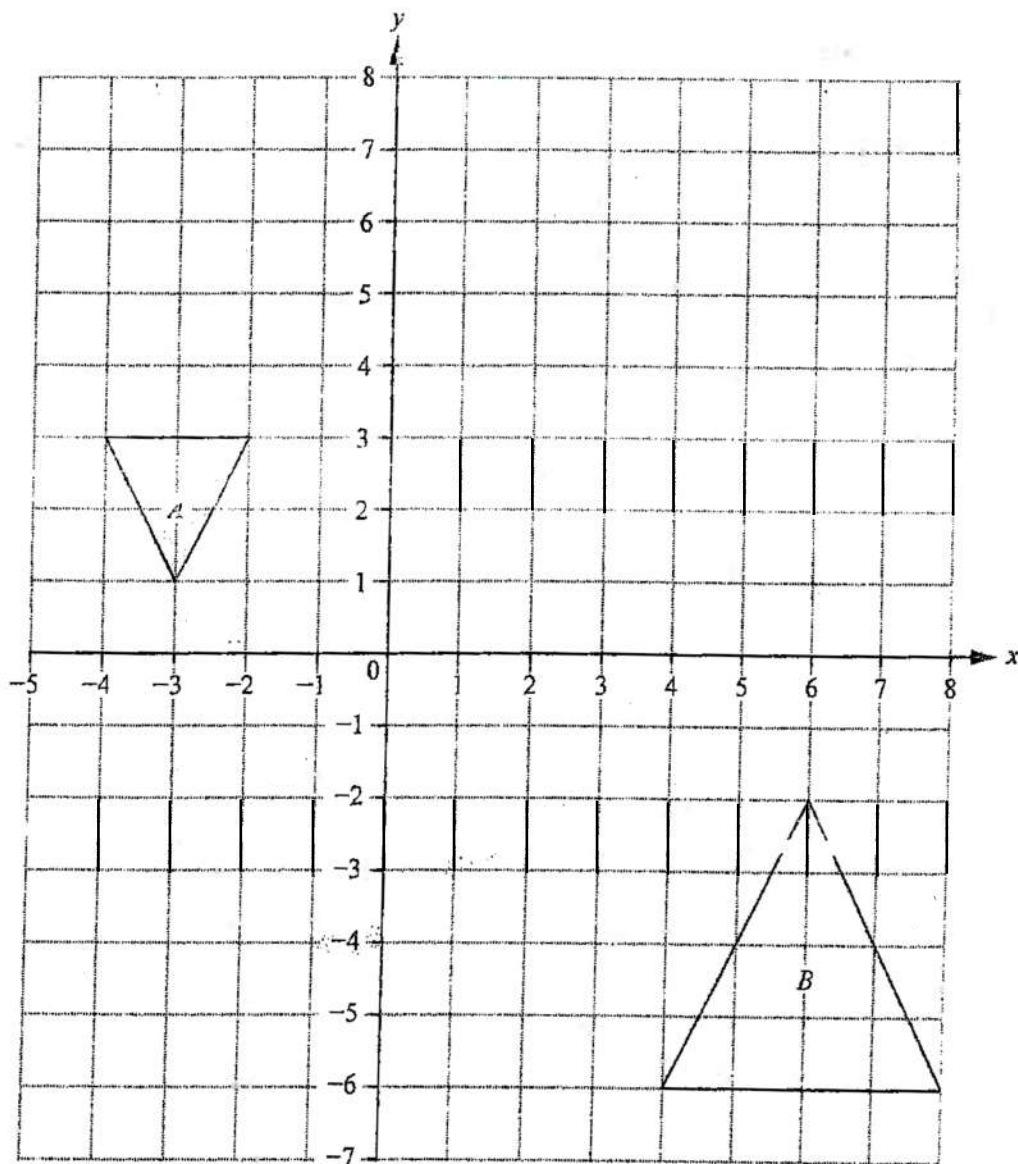
Calculate the time taken to fill the tank.

Give your answer in minutes and seconds.

Answer (c)(ii) min sec [3]



- 8 The diagram shows triangle A and triangle B .



- (a) (i) Reflect triangle A in the x -axis.
Label the image A_1 .

[2]

- (ii) Translate triangle A using the vector $\begin{pmatrix} 3 \\ -1 \end{pmatrix}$

Label the image A_2 .

[2]

- (b) Describe fully the **single** transformation that maps triangle A onto triangle B .

Answer (b)

[3]

- (c) Triangle C has coordinates $(-2, 6)$, $(0, 7)$ and $(0, 5)$.

- (i) On the same grid, draw and label triangle C .

[2]

- (ii) Describe fully the **single** transformation that maps triangle A onto triangle C .

Answer (c)(ii)

[3]



- 10 Tlou took 12 Sesotho tests during one school session.
His marks are shown in the list.

18 12 20 17 17 13 17 10 12 17 13 14

- (a) In how many tests did Tlou score more than 14 marks?

Answer (a) [1]

- (b) Find

- (i) the range,

Answer (b)(i) [1]

- (ii) the mode,

Answer (b)(ii) [1]

- (iii) the median,

Answer (b)(iii) [2]

- (iv) the mean

Answer (b)(iv) [2]

- (c) Tlou sorts his marks into two levels,

pass (less than 14)

and

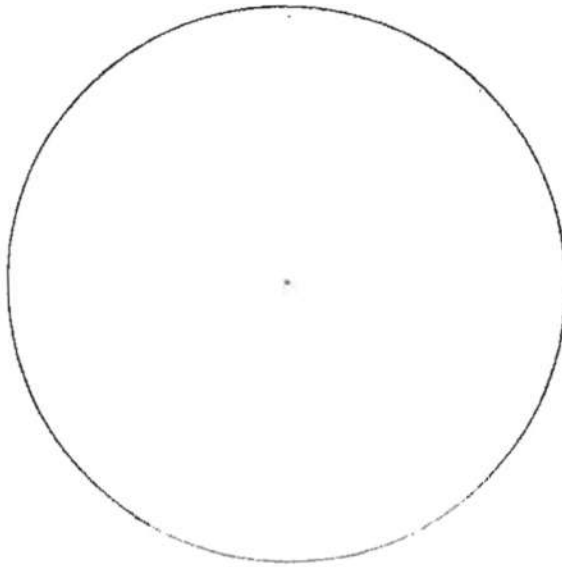
pass with distinction (14 or more).

- (i) Complete the frequency table to show this information.

Level	Pass	Pass with distinction
Frequency		

[1]

- (ii) Complete the pie chart accurately to show this information.
Label each sector.



[3]

- (d) A mark is chosen at random from the list of 12 marks.

Find the probability that this mark is pass with distinction.

Answer (d) [1]

- (e) Tlou took 24 Sesotho tests during the school year.

In how many tests would you expect Tlou to pass with distinction?

Answer (e) [1]

